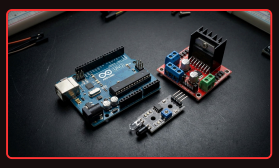
AJ-ORBIT PROJECT BOM

Line Following Robot Component Specification & Bill of Materials

ROBOTICS LAB SYSTEM



Hardware Lineup & Optical Detection Module:
Line follower navigation uses differential steering driven by the Arduino microcontroller, high-current L298N dual bridge, and dual TCRT5000-type infrared reflection sensors tuned for contrast disparity between black path lines and reflective surfaces.

#	COMPONENT REQUIRED	QTY	PRIMARY FUNCTION / SPECIFICATIONS
1	**Arduino Uno Board** MCU	1	ATmega328P master brain; executes line tracking algorithm and drives motor PWM.
2	**L298N Motor Driver Module** POWER	1	Dual H-Bridge module capable of delivering up to 2A per channel to both drive motors.
3	**IR Sensors (Line Tracking)** SENSOR	2	Infrared transmitter & photodiode pair with potentiometer threshold tuning for line sensing.
4	**TT Gear Motors** ACTUATOR	2	Dual-shaft 3-6V DC geared motors providing high starting torque for turns.
5	**Robot Wheels** CHASSIS	4	Rubber-treaded drive and idle stability wheels matched to the TT motor shafts.
6	**18650 Li-ion Batteries** POWER	2	High-drain 3.7V cells in series (7.4V nominal) for stable logic and motor voltage.
7	**Battery Holder** POWER	1	Dual-slot 2x18650 enclosure with pre-soldered leads and switch.
8	**Jumper Wires (M-M & M-F)** HARNESS	1 Set	Standard 20cm multi-colored breadboard wires for sensor signals and motor control lines.
9	**Wood Sheet (Balsa / Plywood)** FRAME	1	Lightweight structural baseplate for mounting the motor brackets, Arduino, and modules.

SENSOR CALIBRATION TIP
Adjust onboard IR trimpots under ambient room light so the status LED illuminates solely over the black line.

SUPPLY ISOLATION
Power motors directly via the 7.4V battery pack through L298N VMS, keeping the Arduino logic supply stable.